

1. Dataset Overview

- The dataset contains 54,673 observations across 625 variables.
- Key variables used for analysis include gender, age, happiness, income decile, and physical activity.
- Top missing variables relate to household and subjective well-being measures.
- Germany (DE), Ireland (IE), and Israel (IL) have the highest respondent counts.
- Gender is fairly balanced (Male: 24,929; Female: 29,727).

2. Exploratory Data Visualizations

- **Country distribution:** Germany, Ireland, and Israel have the largest respondent groups.
- **Gender distribution:** More females than males; negligible missing data.
- **Physical Activity:** Skewed right with a spike at 7 days; a large number are very active.
- **Age:** Wide spread, most respondents are aged between 30 and 70.
- **Happiness:** Bell-shaped distribution skewed toward higher happiness.
- **Income:** Evenly spread, though missing values are significant.

3. Physical Activity Relationships

- **By Gender:** Females are generally more active than males across 7-day frequency levels.
- **By Age Group:** Younger adults (15-44) are more active; activity declines steadily with age.
- **By Country:** Nordic and Western European countries (e.g., DK, SE, FI) show higher average physical activity levels.
- **By Happiness:** Positive relationship; happier individuals tend to be more physically active.

- **By Income:** Mid-income levels (Decile 5-7) show highest physical activity. Extremes (low/high) are less active.